

Syllabus :

PCCS4204	Formal Language and Automata Theory	3-0-0	Credit-3
Module 1 (10 Hours) Alphabet, languages and grammars. Production rules and derivation of languages. Chomsky's hierarchy of languages and Grammars. Regular grammars, regular expressions and finite automata (deterministic and nondeterministic). Closure and decision properties of regular sets. Pumping lemma of regular sets. Minimization of finite automata. Left and right linear grammars. DFA/NFA to regular expression and vice versa using Arden's Formula. Module 2 (10 Hours) Context free grammars and pushdown automata. Chomsky and Greibach normal forms. Parse trees, Cook, Younger, Kasami, and Early's parsing algorithms. Ambiguity and properties of context free languages. Pumping lemma, Ogden's lemma, Parikh's theorem. Deterministic pushdown automata, closure properties of deterministic context free languages.			

Module 3 (10 Hours) Turing machines and variation of Turing machine model, Turing computability, Type 0 languages. Linear bounded automata and context sensitive languages. Primitive recursive functions. Cantor and Gödel numbering. Ackermann's function, μ-recursive functions, recursiveness of Ackermann and Turing computable functions. Module 4 (10 Hours) Church Turing hypothesis. Recursive and recursively enumerable sets. Universal Turing machine and undecidable problems. Undecidability of Post correspondence problem. Valid and invalid computations of Turing machines and some undecidable properties of context free language problems. Time complexity class P, class NP, NP completeness. Text Books: 1. Introduction to Automata Theory, Languages and Computation: J.E. Hopcroft and J.D. Ullman, Pearson Education, 3rd Edition. 2. Introduction to the theory of computation: Michael Sipser, Cengage Learning Reference Books: 1. Automata Theory: Nasir and Srimani, Cambridge University Press. 2. Introduction to Computer Theory: Daniel I.A. Cohen, Wiley India, 2nd Edition.	
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PYQs

2024-25 :

Registration No.

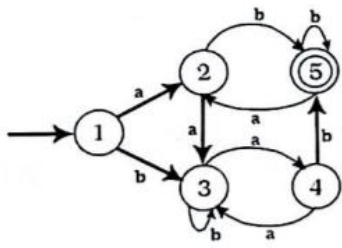


INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG
4th Semester Regular/ Back Paper Examination 2024-2025

Total no. of
Pages: 02

Branch	Computer Science & Engineering	Programme	B. Tech
Subject Name	Formal Language and Automata Theory	Semester	4 th
Subject Code	PCCS4204	Year	2024-25
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.	Maximum Marks	60

Each Question Carry Marks(M) as Indicated in the right-hand Margin

Q's	Questions	M	BL	CO	PO
1.a	Is the language accepted by DFA is always regular? Justify your answer.	01	L1	CO1	PO1
b	Can a PDA have multiple stacks? Explain how it can be possible or not.	01	L1	CO2	PO1
c	Is every NFA also a DFA? Justify.	01	L2	CO3	PO2
d	Can an NFA have more than one initial state? Justify.	01	L3	CO1	PO3
e	Is every regular language also a context-free language? Justify your answer.	01	L4	CO4	PO4
f	Can a language be inherently ambiguous? Explain with a simple example.	01	L5	CO4	PO4
g	Can every CFG be converted to CNF? Explain	01	L5	CO2	PO2
h	Can a Turing Machine simulate any algorithm?	01	L3	CO1	PO3
i	Can any input character be written into PDA stack	01	L4	CO3	PO4
j	Explain the significance of the tape in a Turing Machine. How does it contribute to the computational power of the machine?	01	L5	CO2	PO1
2.a	Find a regular expression corresponding to the language of all strings over the alphabet { a, b } that contain no more than one occurrence of the string 'aa'.	05	L3	CO1	PO2
b	Draw a DFA for the language accepting strings containing neither '00', nor '11' as substring over input alphabets $\Sigma = \{0, 1\}$.	05	L2	CO3	PO3
3.a	Find the regular expression from the given DFA using Arden's theorem 	05	L4	CO1	PO4
b	Let R, S and T be any arbitrary regular expressions. Then, discuss various properties which are true.	05	L5	CO2	PO1
4.a	Construct a CFG for the following language: $L = \{ a^n b^m \mid n \neq m, n \geq 0, m \geq 0 \}$	05	L1	CO4	PO4

b	Show that the grammar $S \rightarrow aS aSb x$ $X \rightarrow Xa a$ where $V = \{S, X\}$ and $T = \{a, b\}$ is ambiguous.	05	L1	CO3	PO3
5.a	Discuss tuples that are defined for Turing Machine with explanation.	05	L2		PO2
b	Design a TM which accepts all strings of the form $a^n b^n$ for $n \geq 1$.	05	L5	CO1	PO3
6.a	Normalise the following grammar into Chomsky's Normal Form(CNF) $A \rightarrow aB bAB$ $B \rightarrow b D$ $D \rightarrow d$ $S \rightarrow aAD \epsilon$	05	L6	CO4	PO3
b	Given the grammar: $S \rightarrow S a S a$ i. Is this grammar left-recursive, right-recursive, or non-recursive? ii. Is this grammar ambiguous?	05	L2	CO4	PO2
7.a	Find the NFA and DFA for the given regular expression using Thompson's set construction method and minimise the DFA $(a + b)^* aa(a + b)^*$	05	L3	CO1	PO2
b	Draw a DFA for the language accepting strings starting and ending with different characters over input alphabets $\Sigma = \{0, 1\}$	05	L5	CO1	PO3
8.	Write short notes (Any two of the following)	05	L1	CO2	PO2
a	PDA vs Turing Machine				
b	Operations on languages	05	L2	CO3	PO3
c	Derivation Tree	05	L3	CO4	PO4
d	Tape alphabet in a Turing Machine?	05	L1	CO1	PO1

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INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG
4th Semester Regular/ Back Paper Examination 2023-24

Total no. of
Pages: 02

Branch	CSE	Programme	B. Tech		
Subject Name	Formal Language and Automata Theory	Semester	4th		
Subject Code	PCCS4204	Year	2023-24		
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.	Maximum Marks	60		
Each Question Carry Marks(M) as Indicated in the right-hand Margin					
Q's	Questions	M	BL	CO	PO
1. a	Write the difference between Finite state machine and PDA.	01	L1	CO1	PO1
b	Is Turing machine a hardware or software?	01	L3	CO4	PO3
c	Discuss various tuples used in a TM.	01	L2	CO3	PO2
d	Give a formal definition of CFG.	01	L4	CO1	PO4
e	Is a 2 stack PDA equivalent to a Turing machine?	01	L1	CO4	PO4
f	Are these given two regular expressions are equivalent? $(a+b)^*$ and $(a^*+b)^*$	01	L3	CO2	PO3
g	What is a right linear grammar? Give example.	01	L4	CO1	PO2
h	Why should we convert NFA to DFA?	01	L5	CO4	PO1
i	Does Chomsky Normal Form (CNF) produce the same language as generated by CFG?	01	L6	CO5	PO4
j	Differentiate between finite automata and Turing machine.	01	L4	CO3	PO3
2. a	Suppose the regular expression is $10 + (0 + 11)0^*1$, where $\Sigma = \{a, b\}$. Construct the NFA for this regular expression then convert it to DFA and then construct the DFA using Thomson's set construction method and optimize the DFA	05	L6	CO3	PO2
b	Draw a DFA for the language accepting odd or even binary numbers strings over input alphabets $\Sigma = \{0, 1\}$?	05	L3	CO4	PO1
3. a	Give context-free grammar that generate the following languages. $\{w \in \{0, 1\}^* \mid \text{the length of } w \text{ is odd and the middle symbol is } 0\}$	05	L4	CO4	PO2
b	Discuss various tuples used in describing PDA. Explain by discussing any PDA of your own.	05	L5	CO5	PO3
c	Obtain a PDA that accepts a language of even palindrome.	05	L6	CO4	PO4
d	Construct a TM for the language $L = \{0^n 1^n 2^n\}$ where $n \geq 1$	05	L5	CO3	PO2

5.a	Reduce the following Grammar into CNF (Chomsky's Normal Form)	05	L4	CO2	PO4
	$S \rightarrow aAD$ $A \rightarrow aB bAB$ $B \rightarrow b$ $D \rightarrow d$				
b	Obtain the Regular expression from the DFA given below using Arden's formula.	05	L2	CO5	PO1
	<pre> graph LR start(()) --> q1((q1)) q1 -- a --> q1 q1 -- b --> q2((q2)) q2 -- b --> q2 q2 -- a --> q3(((q3))) q3 -- a --> q1 style start fill:none,stroke:none </pre>				
6.a	Discuss the hierarchy or Chomsky's classification of grammars with example in each.	05	L3	CO4	PO1
b	Prove that the grammar having the following productions $S \rightarrow aB ab$ $A \rightarrow AB a$ $B \rightarrow Abb b$ Find that the grammar is ambiguous or not.	05	L4	CO3	PO4
7.a	Draw a DFA for the language accepting strings without substring '00' over input alphabets $\Sigma = \{0, 1\}$?	05	L2	CO3	PO2
b	Construct a Turing Machine over $\Sigma = \{0, 1\}$, that contains set of strings of 0's and 1's except those containing the substring 100.	05	L6	CO1	PO3
Write short notes on the following (any Two)					
8.a	Derivation tree	05	L1	CO3	PO1
b	Unrestricted grammar vs CFG	05	L5	CO5	PO3
c	Transition Table used in PDA	05	L3	CO4	PO4
d	Draw DFA that accepts any string which ends with 1 or it ends with an even number of 0's following the last 1. Alphabets are $\{0, 1\}$.	05	L2	CO2	PO1



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Total Number of Pages: 02

B. TECH

Subject Code: PCCS4204

4th Semester Regular/ Back Paper Examination 2022-23

Subject Name: Formal language and Automata theory

Branch: Computer Science & Engineering

Time: 3 Hours

Max Marks: 60

Question Code:

Answer Question No.1 which is compulsory and any five from the rest.

The figures in the right-hand margin indicate marks.

Q1 Answer the following questions: (1x10)

- Write the difference between Finite state machine and PDA.
- Is Turing machine a hardware or software?
- Explain about Arden's formula
- Discuss various tuples of NFA
- What is a dead state?
- Can we find out the regular expression of a DFA having no final state?
- What is a regular set?
- What is the stack symbol in PDA?
- Design a DFA that contains set of all strings ending with 3 consecutive zeroes over the alphabet $\{a, b\}$.
- Can PDA have multiple final states?

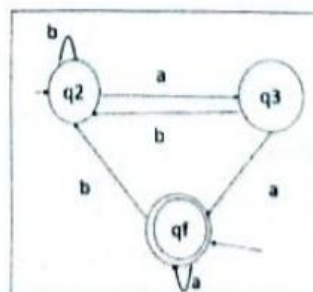
~~Q2~~ a) Suppose the regular expression is $(a|c)^*baa$ where $\Sigma = \{a, b, c\}$. Construct the NFA and optimize the DFA using Thomson's set construction method. (5)

b) Design a DFA to read strings made up of letters 'CHARIOT' and recognize these strings that contains the word 'CAT' as a substring. (5)

~~Q3~~ a) Obtain a PDA that accepts a language of odd palindrome. (5)

b) Write Regular expression for the following $L = \{a^n b^m : m, n \text{ are even}\}$
 $L = \{a^n, b^m : m \geq 2, n \geq 2\}$ (5)

~~Q4~~ a) Convert the following DFA to Regular Expression using Arden's formula. (5)





- b) Design a PDA to accept all strings to accept of the form $\{a^n b^n \text{ where } n \geq 1\}$ (5)

Q5 ✓ a) Construct a Turing Machine for the language $L = \{0^n 1^n\}$ where $n \geq 1$ (5)

- b) Design a DFA using intuition method, which accepts set of all strings containing all strings having mod 3 is zero where string is treated as binary numbers. (5)

Q6 ✓ a) Write the formal definition of Turing machine, explaining all tuples. (5)

- b) Define a Finite State Machine (FSM). What are the elements of an FSM? What is a universal state in an FSM? (5)

Q7 ✓ a) Obtain a PDA that accepts a language of even palindrome. (5)

- b) Obtain DFAs to accept strings of a's and b's having exactly one a. (5)

Q8 ✗ **Write short notes on any two of the following** (5x2)

- a) Regular grammar
- b) NFA
- c) Construct NFA for regular expression $V = (01+10)$.
- d) Limitations of the Turing machine?

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B. TECH

Subject Code: PCCS4204

4th Semester Regular/Back Paper Examination 2021-22
Subject Name: Formal language and automata theory
Branch: Computer Science & Engg.

Time: 3 Hours
Max Marks: 60

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right-hand margin indicate marks.

(1x10)

Q1

Answer the following questions:

- a) ✓ What is a CFG.
- b) ✓ What is a finite state machine?
- c) ✓ What is a recursive grammar?
- d) ✓ Give the formal definition of a CFG.
- e) ✓ What is PDA?
- f) ✓ What are the Hierarchy of grammar?
- g) ✓ What is a regular expression?
- h) ✓ What is a non-recursive grammar?
- i) ✓ Design a DFA that contains set of all strings ending with 3 consecutive zeroes over the alphabet {a,b}.
- j) ✓ Define a regular set with example.

Q2

a) ✓ Design the DFA from the regular expression $(a|b)^*baa$. Optimize the DFA. (5)

b) ✓ Discuss the elements of a TM. (5)

Q3

a) ✓ Obtain a PDA that accepts a language of odd palindrome. (5)

b) ✓ Design a TM that erases all non-blank symbols on the tape over the alphabet {a, b}. (5)

Q4

a) ✓ Design a TM which accepts all strings of the form $\{a^{2n}b^{2n} \text{ where } n \geq 1\}$. (5)

b) ✓ Design a PDA to accept all strings to accept of the form $\{a^n b^n \text{ where } n \geq 1\}$ (5)

Q5

a) ✓ Define a TM. Discuss the Tuple of the Turing Machine defining all the Tuples. (5)

b) ✓ Design a DFA which accepts set of all strings containing all strings having mod 3 is zero where string is treated as binary numbers. (5)

Q6

a) ✓ What are the various operations on strings? Discuss with examples. (5)

b) ✓ Define a Finite State Machine (FSM). What are the elements of an FSM? What is a universal state in an FSM? (5)

- a) Obtain a PDA that accepts a language of even palindrome.
- b) What is the difference between FSM and PDA? Discuss with example.

Write short notes on any two of the following

- a) Regular grammar
- b) NFA
- c) Context Free Grammar
- d) Right Derivation of a string